ID	Name	Description
FR-01	Account Registration	The system shall allow a Guest to create an account by submitting username, email and password.
FR-02	Required Field Validation	The system shall reject registration requests that omit any required field and shall return validation errors identifying each invalid or missing field.
FR-03	Uniqueness Enforcement	The system shall reject registration if the submitted username or email is already in use.
FR-04	Credential Creation	Upon successful registration submission, the system shall create exactly one authentication record for the new user and securely store the password as a hash (not plaintext).
FR-05	Pending Verification State	Upon successful registration submission, the system shall set the new user's status to PendingVerification until email verification is completed.
FR-06	Verification Token Generation	The system shall generate a unique email verification token associated with the newly registered user.
FR-07	Verification Email	The system shall send a verification email to the user's registered email address containing a verification link that includes the token.
FR-08	Verify Email	The system shall provide a verification operation that accepts a verification token and activates the account if the token is valid and not expired.
FR-09	Activate Account	When verification succeeds, the system shall set the user status to Active and record the verification completion timestamp.
FR-10	Token Failure Handling	If the verification token is invalid, expired, or already used, the system shall reject verification and return an error message indicating the reason.
FR-11	Resend Verification	The system shall allow a user in PendingVerification status to request a new verification email. Resending generates a new token and invalidates prior tokens, or clearly defines the rule.
FR-12	Login Restrictions	The system shall prevent users with status Suspended or Deleted. or locked out from authenticating successfully.
FR-13	Login Operation	The system shall allow a Guest to submit email and password to request authentication.
FR-14	Credential Lookup	Upon a login request, the system shall locate the user account using the submitted email.
FR-15	Status-Based Access Control	The system shall deny authentication if the user's status is not Active.
FR-16	Lock State Enforcement	The system shall deny authentication if the account is locked and the current time is earlier than lockedUntil.
FR-17	Password Verificaiton	If the user is eligible to authenticate (Active and not locked), the system shall validate the submitted password against the stored passwordHash.
FR-18	Failed Login Attempt Tracking	If password verification fails, the system shall increment failedLoginAttempts for the associated UserCredentials record.
FR-19	Lockout Threshhold	If failedLoginAttempts reaches a configured threshold, the system shall lock the account.
FR-20	Succesful Login Reset	If password verification succeeds, the system shall reset failedLoginAttempts and clear locked/lockedUntil if applicable.

FR-21	Session Issuance	Upon successful authentication, the system shall create an authenticated session (or issue an authentication token) for the user.
FR-22	Failed Login Error Messaging	If login fails due to invalid credentials, the system shall return a generic failure message without disclosing whether the email exists.
FR-23	Lockout Messaging	If login fails due to lockout, the system shall inform the user that the account is temporarily locked and provide option to reset password.
FR-24	Last Password Update Time	The system shall maintain passwordUpdatedAt when the password is changed.
FR-25	Create Channel Operation	The system shall allow an authenticated user to create a Channel by submitting a name, and optionally a description and an avatar image.
FR-27	Owner Assignment	Upon successful channel creation, the system shall associate the new Channel with exactly one owning User.
FR-29	Initialize Subscriber Count	Upon channel creation, the system shall iniitalize subscriberCount.
FR-30	Update Audit Fields	The system shall set createdAt at channel creation time and set updatedAt when channel details are modified.
FR-31	Update Channel Profile	The system shall allow the channel owner to update channel profile fields such as name, description, and avatarUrl.
FR-33	Multiple Channels	A user shall have the ability to create multiple channels under their User account.
FR-34	Upload Video Operation	The system shall allow an authenticated user to upload/publish a Video to a specified Channel by submitting a title, the video file and optionally description and thumbnail image.
FR-38	Set Audit Timestamps	Upon video creation, the system shall set createdAt. Upon any edit to video metadata, the system shall set updatedAt.
FR-39	Edit Video Metadata	The system shall allow the video's owning channel owner to update video metadata, including title, description, thumbnailUrl, and visibility.
FR-41	Enforce Private Visibility Access	If a video's visibility is Private, the system shall restrict access so that only the owning User can view the video.
FR-42	Start Playback Operation	The system shall allow a user to request playback of a Video.
FR-43	Enforce Video Visibility	Before starting playback, the system shall verify that the requesting user is authorized to view the video based on visibility: - Public videos are viewable by authorized guests and authenticated users. - Private videos are viewable only by the owning user and Admins.
FR-44	Playback Delivery	For guests and authorized users, the system shall initiate video playback and provide the necessary stream/playback information to the client.
FR-45	Membership Evaluation for Playback Features	Before playback begins, the system shall determine the user's current membership benefits to determine: - Wheter ads should be shown. - Wheter HDR playback is allowed.

FR-46	Ad Display Logic	The system shall display advertisements during playback for users whose membership is not ad-free, and shall not display advertisements for users whose membership includes ad-free benefits.
FR-47	Skip Ads for Paid Members	If the user's membership is ad-free, the system shall not display ads during playback.
FR-50	Update View Event	When playback ends (or the client reports completion), the system shall record: - watchDuration - completed (true if fully watched, otherwise false) - watchedAt timestamp for the ViewEvent
FR-51	Handle Partial Playback	If playback is interrupted, the system shall still record a ViewEvent withand the partial watch duration.
FR-52	Playback Denial Behavior	If the user is not authorized to view the video (e.g., private video and not owner/admin), the system shall deny playback and shall not mark the video as completed.
FR-53	React Operation	The system shall allow an authenticated user to create a Reaction of type Like or Dislike for a target that is either: - a Video (videoId provided), or - a Comment (commentId provided).
FR-56	Visibility and Access Enforcement	The system shall deny reactions to content that the user is not authorized to view (e.g., Private video not owned by the user).
FR-57	One Reaction Per User Per Target	The system shall ensure a user has at most one active Reaction per target (one per Video or one per Comment at a time).
FR-58	Reaction Update Behaviour	If a user already has a Reaction on the target, the system shall update the existing Reaction to reflect the new type (switch Like to Dislike) or clicking the same reaction removes it.
FR-59	Reaction Timestamp	The system shall record createdAt for a newly created Reaction (and update timestamp if you choose to track edits).
FR-60	Update Reaction Counts on Comments	When a Reaction is created/updated/removed on a Comment, the system shall update the Comment's likeCount and dislikeCount accordingly.
FR-61	Notify Content Ownwer on Reaction	When a Reaction is successfully applied to a Video or Comment, the system shall create a Notification for the content owner (recipient user) with: - type indicating Like(Video) or Like(Comment) - actorUserId set to the reacting user - sourceType and sourceId referencing the reacted content
FR-62	Do Not Notify Self	The system shall not generate notifications when the actor user is the same as the recipient user.
FR-63	Reaction Response	The system shall return the resulting reaction state to the user (e.g., Like applied, switched to Dislike, removed) along with any updated counts relevant to the target.
FR-64	Post Comment Operation	The system shall allow an authenticated user to create a Comment on a Video.
FR-65	Authorization Enforcement (Comment)	The system shall deny comment creation requests from unauthenticated users.

FR-67	Comment Content Record	Upon successful validation and authorization, the system shall create a Comment linked to: - the authoring User (userId) - the target Video (videoId) - and optionally a parent Comment (parentId) when the comment is a reply.
FR-68	Reply Comment Operation	The system shall allow an authenticated user to reply to an existing Comment.
FR-69	Parent Comment Validation	When creating a reply, the system shall verify the parent Comment exists and is associated with the same Video.
FR-70	Comment Status on Creation	The system shall set newly created comments to an Active status by default, unless moderation rules require otherwise.
FR-71	Edit Comment	The system shall allow the comment author to update comment content, and the system shall set updatedAt when edited.
FR-73	Notify on Comment	When a new top-level comment is created on a video, the system shall create a Notification for the Video owner with: - type = Comment(video) - actorUserId = commenting user - recipientId = video owner - sourceType = Comment - sourceId = new comment id
FR-74	Notify on Reply	When a reply comment is created, the system shall create a Notification for the parent comment's author with: - type = Reply(comment) - actorUserId = replying user - recipientId = parent comment author - sourceType = Comment - sourceId = reply comment id
FR-75	Do Not Notify Self (Comment)	The system shall not create a Notification when the actor user is the same as the recipient user.
FR-76	Comment Response	Upon successful comment creation or reply, the system shall return the created comment identifier and the displayed comment data (time since comment posted).
FR-77	Create Playlist Operation	The system shall allow an authenticated user to create a Playlist by submitting name, visibility (Public/Private) and optional description.
FR-79	Playlist Ownership	Upon creation, the system shall associate the Playlist with exactly one owning User.
FR-81	Update Playlist Metadata	The system shall allow the playlist owner to update playlist metadata and shall set updatedAt when modified.
FR-83	Add Video to Playlist	The system shall allow the playlist owner to add a Video to a Playlist, creating a PlaylistVideo association.

FR-85	No Duplicate Videos on Playlist	The system shall prevent the same Video from being added to the same Playlist more than once at a time.
FR-86	Maintain Playlist Ordering	When adding a Video, the system shall assign a position in the playlist such that each PlaylistVideo in a playlist has an order.
FR-87	Remove Video From Playlist	The system shall allow the playlist owner to remove a Video from a Playlist by deleting the associated PlaylistVideo record.
FR-88	Repack Positions After Removal	After a removal, the system shall update remaining PlaylistVideo positions to preserve a valid, continuous ordering according to the platform's ordering rule.
FR-89	Reorder Playlist	The system shall allow the playlist owner to reorder videos in a playlist by submitting the desired ordering.
FR-90	Validate Reorder Request	The system shall reject reorder requests that reference videos not in the playlist, omit existing items, or contain duplicates.
FR-91	Persist Reorder Results	Upon successful reorder, the system shall update PlaylistVideo position values to reflect the new ordering.
FR-92	Playlist Visibility Enforcement	If a Playlist's visibility is Private, the system shall restrict access so that only the playlist owner and Admins can view the playlist contents.
FR-93	Subscribe Operation	The system shall allow an authenticated user to subscribe to a Channel.
FR-95	Channel Existence Validation	The system shall reject subscription requests if the target Channel does not exist.
FR-96	No Self-Subscription	The system shall deny a subscription request if the requesting user is the owning user of the target Channel.
FR-97	Single Active Subscription	The system shall ensure that a user may have at most one Active Subscription to the same Channel at any given time.
FR-98	Create or Reactivate Subscription	When a user subscribes: - If no prior Subscription exists, the system shall create a new Subscription Active status and set date of creation. - If a prior Subscription exists with status Cancelled, the system shall reactivate it by setting status Active.
FR-99	Unsubscribe Operation	The system shall allow an authenticated user to unsubscribe from a Channel.
FR-100	Cancel Subscription	Upon unsubscribe, the system shall set the Subscription status to Cancelled, preserving history.
FR-101	Subscriber Count Consistency	The system shall maintain the Channel subscriberCount such that it reflects the number of Active Subscriptions and shall update the count when subscriptions are activated or canceled.
FR-102	Subscriber Count Update	When a subscription becomes Active or Cancelled, the system shall update the channel's subscriberCount accordingly.
FR-103	Subscription Response	The system shall return the resulting subscription state (Active/Cancelled/Already Active) to the user.

FR-104	View Membership Plans	The system shall allow an authenticated user to view available MembershipPlans, including name, price, and feature flags, limited to plans where that are active.
FR-106	Select MembershipPlan	The system shall allow an authenticated user to select a MembershipPlan and request a membership change.
FR-107	Validate Plan Availability	The system shall reject membership changes if the selected MembershipPlan does not exist or is not active.
FR-108	Create Membership Record	When a membership change is approved, the system shall create a new UserMembership record linked to the User and the selected MembershipPlan.
FR-109	End Prior Active Membership	If the user already has an Active membership, the system shall end the prior membership.
FR-110	Single Active Membership	The system shall ensure that a user has at most one UserMembership with status Active at any given time.
FR-111	Cancel Membership	The system shall allow an authenticated user to cancel their current membership by setting the active UserMembership status to Canceled and setting end date.
FR-112	Membership Benefit Exposure	The system shall make the user's effective membership benefits available to relevant operations.
FR-113	Membership Change Response	The system shall return the resulting membership state (Active/Canceled/Expired), the effective plan, and relevant timestamps after a membership change.
FR-114	Authenticated Access Control	The system shall require users to be authenticated before performing protected operations including channel creation, video upload, reactions, commenting, playlist management, subscriptions, and membership changes.
FR-115	Ownership Authorization Enforcement	The system shall enforce ownership-based authorization such that only the resource owner may modify or delete that resource. All content shall be visible to admin/moderators and have the ability to remove content.